



Plant Patrol

A platform to help managing greenhouses for a plant store, on an eco-friendlier way.

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IES-LEI

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Problem

- Difficult to monitor plant health, growing conditions, and inventory manually
- Human error or lack of resources can lead to plant loss and resource waste.
- Plants may suffer from poor ventilation, incorrect watering, or insufficient light exposure.

Consequences

- Increased financial losses
- Decreased customer satisfaction
- Difficulty scaling the business as demand grows



PlantPatrol: The Solution



System that uses real-time sensors to monitor plant conditions (e.g., temperature, humidity, light exposure, air quality)



Integrates an inventory management system to keep track of plant stock and resources.



Provides a customer-facing catalog for browsing plant details and availability.

Comparison to Existing Solutions

	Environmental Monitoring	Inventory Management	Customer Interaction	Catalog for Customers
Sirin Software Greenhouse Monitoring	✓	✗	✗	✗
Growflo	✗	✓	✗	✗
Renkeer Greenhouse Monitoring	✓	✗	✗	✗
PlantPatrol	✓	✓	✓	✓

User Stories

As a <user role>

I want <goal>

so that <benefit>.



As a Manager

I want to control the plant's status

so that I can see if everything is good, and issue an alert to the employees if something is wrong

Acceptance Criteria:

- The system allows to see real-time data about the sensors
 - The system allows you to issue an alert to employees about the care that should be taken for a given group
-

As a Manager

I want to see historic statistics about what the sensors have recorded,
so that I can see if the systems are well configured

Acceptance Criteria:

- The system allows you to view graphs showing the atmospheric conditions for each group of plants



As a Manager

I want to check up the inventory

so that I can know what to order in advance

Acceptance Criteria:

- The system allows you to define the minimum quantity of a product
 - A notification is sent to inform you that the stock of a certain product is running low
-

As a Manager

I want to add and manage the employees accounts on the system

so that a new hired employee can access the system

Acceptance Criteria:

- Authentication system
- The system allows you to change passwords
- The system allows you to manage information of each employee

As an Employee

I want to receive a notification if a greenhouse sensor detect something abnormal
so that I can adjust the systems

Acceptance Criteria:

- Push notification system

As an Employee

I want to setup the watering system based on sensor data
so that I can manage the conditions easily and save water and energy

Acceptance Criteria:

- The system allows to see real-time data about the sensors
- The system allows you to define the conditions to which a group is subject to (amount of water supply, etc.)



As an Employee

I want to update the stock of plant's

so that the client and manager can access the current plants available

Acceptance Criteria:

- The system allows you to change the quantity of each product.
 - The system allows you to create and delete plants.
-

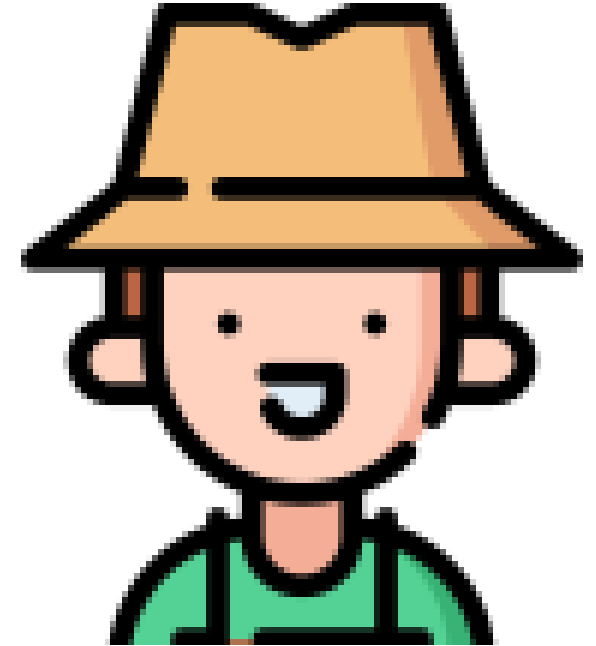
As an Employee

I want to be able to help previous clients with their plant issues

so that I have happy customers and a good plant support

Acceptance Criteria:

- The system has a chat for a plant support line that allows the employees to speak and help customers.





As a Client

I want to see the available plants on the store
so that I can know when to buy and how many to buy

Acceptance Criteria:

- The system allows you to check whether a given plant is available or not
 - Push notification system that notifies when a plant is available again
-

As a Client

I want to be able to talk with the employees through a chat
so that clarify my doubts about issues that may occur

Acceptance Criteria:

- The system has a chat for a plant support line



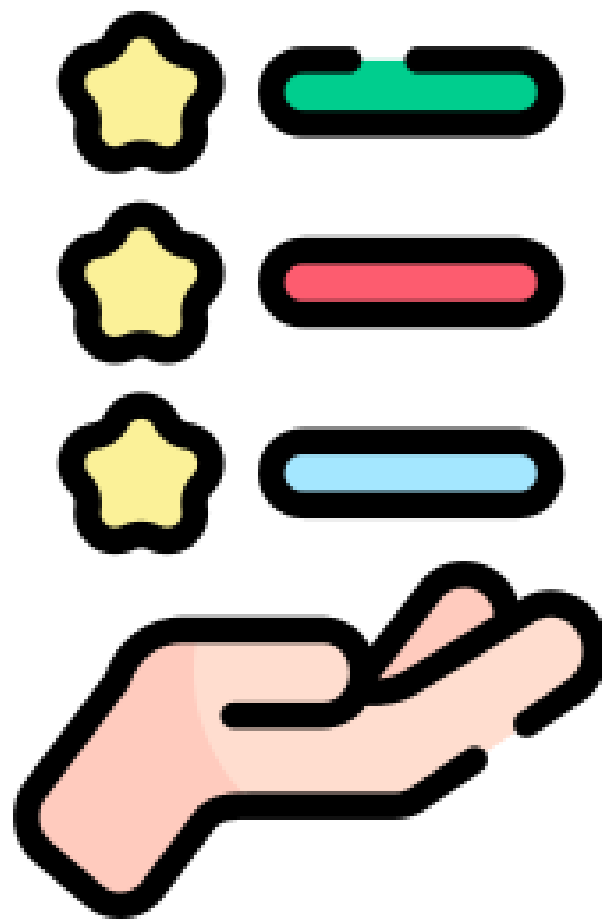
As a Client

I want to know which plants have certain characteristics
so that I know which will do best in my garden.

Acceptance Criteria:

- The system allows you to check the ideal characteristics for each plant

Features





Manager:

- Manage all the employees
- Get passwords for the new ones
- Create a greenhouse and associate micro-controllers
- Create a new alert
- Create rules and edit them

Employee:

- Edit and Add a plant in the inventory
- Associate micro-controllers
- Chat system so that an employee can help clients in need
- Create rules and edit them



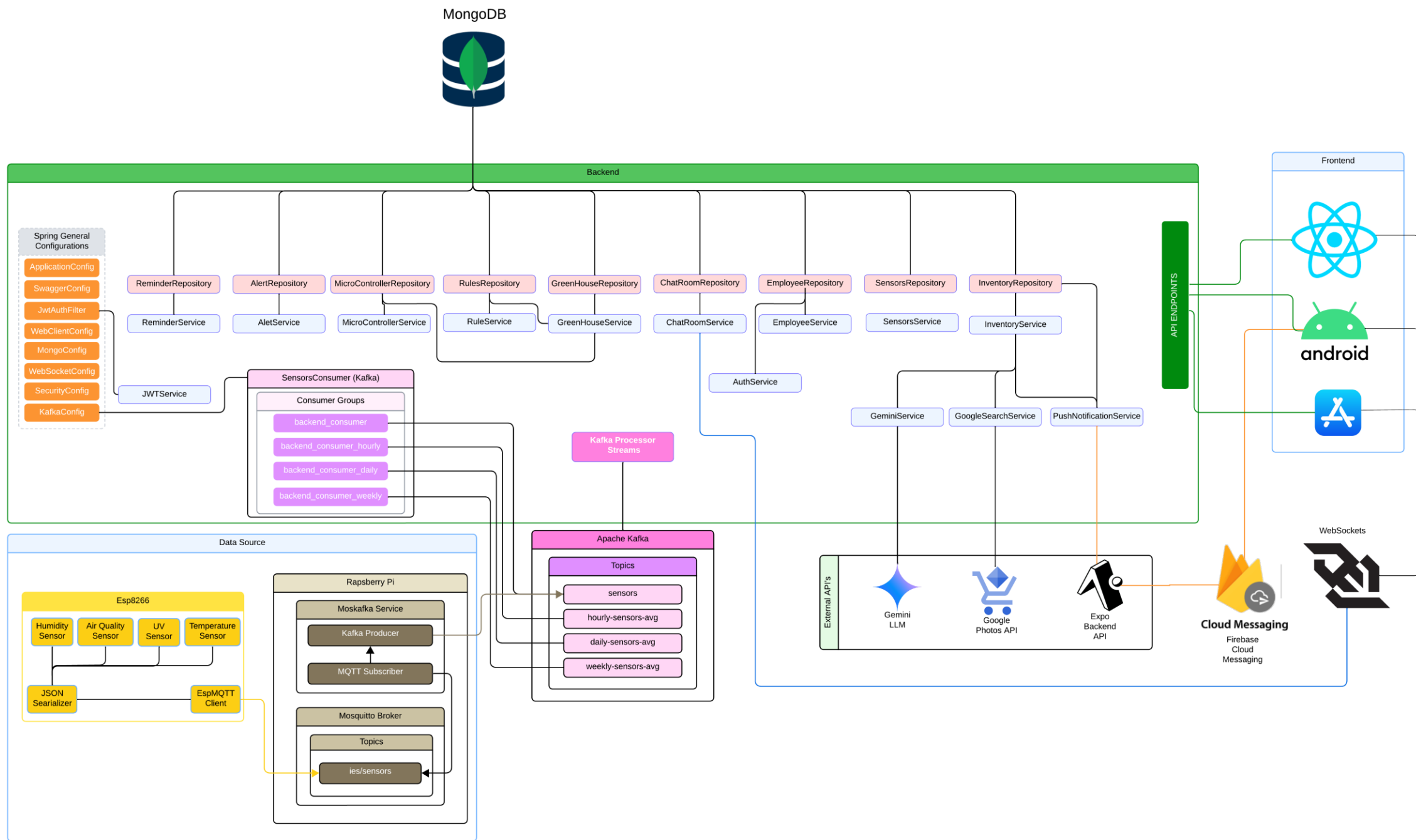
Client:

- Chat system so that a client can get help from an employee
- See if a plant is available and mark them
- A client receives a notification when a plant that was marked is available



Arquitecture







Next Iteration

iteration:@next

Title		...	Assignees	...	Status	...	Story Points	...	Priority	...	Iteration	...
1	Implement TLS connection	#194		▼	Todo	▼				▼	Iteration 6	▼
2	Authentication MQTT Broker	#195		▼	Todo	▼				▼	Iteration 6	▼
3	Loading Component	#193		▼	Todo	▼				▼	Iteration 6	▼
4	Add the option to delete an alert	#196		▼	Todo	▼				▼	Iteration 6	▼
5	Implement employee event logging	#197		▼	Todo	▼				▼	Iteration 6	▼
6	Add software monitoring systems	#200		▼	Todo	▼				▼	Iteration 6	▼

+ You can use Control + Space to add an item



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